Research 2026-10-21: workspace UI patterns, invitation email, Gemini Deep Research, file hand-over

Method: three sourced web-research passes. "UNVERIFIED" = not confirmed from a fetched page. Numbers for hover pills are our own choice, not copied from any product.

A. Breadcrumbs, lists, sidebar, search, menus

Breadcrumbs

- W3C APG: no special keyboard model; a `nav` landmark with `aria-label`, `ol > li > a`; last item `aria-current="page"` (https://www.w3.org/WAI/ARIA/apg/patterns/breadcrumb/).
- Carbon: never wrap; when short show first + last two, middle in an overflow menu; phone: overflow + one crumb (https://carbondesignsystem.com/components/breadcrumb/usage/). Atlassian: collapsed items = ellipsis labelled "Show more breadcrumbs"; apps with a sidebar may drop the top level (https://atlassian.design/components/breadcrumbs/usage). NN/g: show location, not history; current page last and unlinked (https://www.nngroup.com/articles/breadcrumbs/).
- Notion shows sibling pages when hovering a crumb (third-party source only). Supabase moved to one organization at a time with an org picker in the header and an `/organizations` page (https://supabase.com/changelog/33670-upcoming-breaking-change-to-dashboard-navigation). Linear/GitHub/Figma/Drive crumb behaviour: UNVERIFIED.
- Rules adopted: crumb items are link + small chevron button that opens a sibling list (click, not hover only); middle crumbs hidden on narrow screens; link height >= 24px.

Org / project list pages

- Vercel: grid/list toggle saved per user, sort by activity/name, find bar, "Add New" (https://vercel.com/docs/projects/managing-projects). Delete = typed project name; "Pause" offered as a softer option.
- Supabase roles Owner/Administrator/Developer/Read-only; invites expire in 24h, can be revoked and resent; an org needs at least one owner (https://supabase.com/docs/guides/platform/access-control). GitHub org delete: typed name, irreversible warning, alternatives offered (https://docs.github.com/en/organizations/managing-organization-settings/deleting-an-organization-account).
- NN/g confirmations: name the item and consequences, label buttons by outcome, prefer undo, typed confirmation only for rare high-stakes actions (https://www.nngroup.com/articles/confirmation-dialog/). Carbon empty states: first use / no results / error; title, body, action (https://carbondesignsystem.com/patterns/empty-states-pattern/).
- Rules adopted: one toolbar (search, sort, view toggle persisted, one primary create button); three empty-state kinds; invitations always reachable (header button on org and project pages, command palette, and the pending-invite entry); danger zone at the bottom of Settings; soft delete first with restore, typed name for permanent delete.

Sidebar: Notion sections Recents/Favorites/Teamspaces, each collapsible, Trash/Help pinned at the bottom (https://www.notion.com/help/sidebar). Linear: hide rare items behind "More", counts only for actionable things (https://linear.app/changelog/2024-12-18-personalized-sidebar). Rule: sidebar = navigation inside the current project; account, org settings and members live elsewhere; badges only for actionable counts; collapsible sections with remembered state.

Search / palette: GitHub prefixes `> # @ / ?`, scope chip (https://docs.github.com/en/get-started/accessibility/github-command-palette); Linear/Notion show recents on empty query and have filters (https://linear.app/docs/search, https://www.notion.com/help/search); Vercel palette is desktop-only so a mobile search button is needed. APG list-autocomplete combobox (focus stays on input, `aria-activedescendant`). Rule: ONE search entry in the header; groups Actions / Navigate / Documents; prefixes; filter chips; recents.

Account menu / header actions: Supabase keeps the account menu top-right; Claude/Notion/Linear keep it in the sidebar. Claude's exact item list UNVERIFIED. Rule: menu width 220-280px (not the sidebar width), rows 28-32px; items: name+email, settings, help, shortcuts, release notes, backup, sign out. Header right: Search, notifications bell (review inbox), pin, copy link, overflow "…" with page actions.

Density: Primer ActionList `inset` variant; WCAG 2.2 target size >= 24x24 (https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum.html). One rule set: row height 30px, radius 7px, padding 4px 8px, inset from container edges, focus by outline.

Dropdowns: `appearance: base-select` — Chrome/Edge 135+, Safari 27 (Sep 2026), Firefox behind a flag, so not Baseline (https://web.dev/blog/web-platform-09-2026). We use our own select-only combobox (APG pattern: https://www.w3.org/WAI/ARIA/apg/patterns/combobox/examples/combobox-select-only/) so look and behaviour are identical everywhere.

B. Invitation email (research, no code yet)

- Supabase built-in SMTP: 2 messages/hour and only to addresses of your Supabase org team, all plans (https://supabase.com/docs/guides/auth/auth-smtp). Not usable.
- Resend free: 100/day but `onboarding@resend.dev` only sends to your own account address; a verified domain is needed (https://resend.com/docs/knowledge-base/403-error-resend-dev-domain). Brevo (300/day) cannot authenticate gmail.com senders; Postmark rejects Gmail sign-ups; SendGrid has no permanent free plan; SES sandbox = verified recipients only.
- Gmail SMTP with an app password: works for any recipient, mail is signed by Google (SPF/DKIM/DMARC aligned), about 500/day (third-party figure). Needs 2-Step Verification + a 16-character app password. Supabase Edge Functions block outgoing ports 25 and 587 (https://supabase.com/docs/guides/functions/limits); 465 UNVERIFIED. A Vercel Node function with nodemailer on 465 is the planned path (UNTESTED until credentials exist).
- Google Apps Script web app (`MailApp.sendEmail`): HTTPS only, no app password; consumer cap 100 recipients/day; URL is public so a shared secret is mandatory.
- Domain route later: ~10 USD/year + Resend free tier gives a branded sender.
- Animation support (caniemail, data updated 2026-09-16): CSS keyframes only in Apple Mail (macOS/iOS), Samsung, Thunderbird, HEY, Fastmail; NOT Gmail, any Outlook, Proton. Animated GIF plays in Gmail/Apple/Yahoo/Outlook.com but not Outlook for Windows (shows a frame). Base64 images are blocked in Gmail, inline SVG and video unsupported, AMP needs a registered non-Gmail sender. Gmail clips HTML above ~102KB; `<style>` only in `<head>`, 16KB.
- Technique adopted: static inline-styled 600px table layout (works everywhere) + hosted animated GIF hero (Gmail etc.) + optional head `<style>` keyframes for Apple Mail/iOS/Samsung + dark-mode overrides + bulletproof button + plain-text part + alt text. Base styles stay in the finished state so clients that strip animation still show the complete design.
- Security: scanners (Microsoft Safe Links) open links, so the link must only load a harmless page; the token is consumed by the "Accept" click (already how our `#/join/<token>` works). Never put the token in preview text. Rate-limit invitations; escape inviter/org names; set Reply-To to the inviter.

- Copy pattern: subject "{Inviter} invited you to join {App}", one headline, one button, plain URL under it, "expires in N days", "If you weren't expecting this, ignore it — nothing happens until you click". GitHub and Vercel use 7-day expiry (https://docs.github.com/en/organizations/managing-membership-in-your-organization/inviting-users-to-join-your-organization).
- Owner action unavoidable: a sender credential (Gmail app password, or Apps Script deploy) must be created by the owner; it goes into the Vercel environment settings, never into chat or code.

C. Gemini Deep Research (Oct 2026)

- "Argon" = Gemini 4 Argon, Google's flagship reasoning model, thinking levels low/medium/high (https://ai.google.dev/gemini-api/docs/models/gemini-4-argon, updated 2026-10-14). "Argon High" as a Deep Research mode/UI label is UNVERIFIED — most likely Argon at the High thinking level; the owner should send a screenshot of the picker.
- Plans (https://gemini.google/subscriptions/): AI Plus 4.99, AI Pro 19.99, AI Ultra from 99.99 USD/month. Limits are compute-based, refresh every 5 hours with a weekly cap (https://support.google.com/gemini/answer/17004136); exact Deep Research counts and parallel-run limits UNVERIFIED.
- Files: "Add Files"; up to 10 files per prompt, 100MB each, ZIP up to 10 files (https://support.google.com/gemini/answer/14903178). Whether .md/.txt are accepted by Deep Research UNVERIFIED -> convert to PDF or Google Docs. Google Search on by default; Drive/Gmail optional. The plan can be edited before "Start research"; export to Docs.
- Prompt practice (Google API doc + guides): state objective, scope, time frame, structure of the output; name preferred source types; ask for inline citations with URLs and for "unknown" instead of estimates; say which attached file is authoritative; edit the plan before starting; split big topics into 3-6 runs and synthesise. Template in docs/11.

D. Why files from the cloud session often do not open

- Cloud sessions have a view-only file pane; edits only inside the working directory. Official docs do not describe a download action (https://code.claude.com/docs/en/desktop.md). Our session rules say the app opens only files inside the primary working directory (/home/user/gf), the scratchpad and memory directories.
- Earlier notes were written to /home/user (outside /home/user/gf) -> the likely reason they did not open.
- Reliable routes: files inside /home/user/gf committed and pushed (GitHub link), private Artifact pages (.html/.md, 16 MiB max), Claude Docs, teleport. Plan: put every deliverable in /home/user/gf/docs, push to GitHub, and publish a private Artifact for reading.

Research 2026-10-21: planning layer (master file > phases > steps), decisions, document health

Pages fetched 2026-10-21. DESIGN = design inference by the researcher, no authoritative standard. UNVERIFIED = not confirmed. Feeds master plan phase 4.

What other tools do

- Notion: sub-items by parent relation; deleting a parent deletes children; dependencies are an opt-in feature [N1]. Linear: initiatives group projects; statuses Proposed/Planned/Active/Completed/Canceled; health is a manual on/at-risk/off-track update [L1]; project graph counts unestimated issues as 1 point, in-progress at 1/4 in forecasts, cancelled issues lower scope [L3]. GitHub: sub-issues up to 8 levels and 100 children; Markdown tasklist blocks retired [G1][G2]; projects have typed fields, table/board/roadmap [G3].
- Obsidian Tasks: status = one character, each with a type TODO/IN_PROGRESS/ON_HOLD/DONE/CANCELLED/NON_TASK [O1]; Dataview `completed` only for `x`, `fullyCompleted` also needs all subtasks [O2]. Org-mode: `[/]`/`[%]` count direct children by default; parent box derives from children [OM1].
- Shape Up: hill charts track scopes, not tasks; percent complete misleads because tasks keep appearing; appetite = fixed time, variable scope; pitch = Problem, Appetite, Solution, Rabbit holes, No-gos [B1][B2].
- Takeaway: every tool separates a typed row (status, dates, parent) from free text. Use rows as the source of truth and render Markdown from them; import/export checkbox Markdown through a status-character map. We need states beyond done/not done: blocked, cancelled, deferred.

Recommended schema (DESIGN; reuses docs, doc_revisions and the AI change sets)

```
plan_nodes: id, project_id, ref (step:4-3, unique per project), kind (phase|step; sub-
step = step with a step parent),
  parent_id fk on delete RESTRICT, position text COLLATE "C" (fractional key)
unique(parent_id, position),
  title, notes_md, doc_id null, status (todo|doing|done|blocked|cancelled|deferred),
weight int>=0 default 1,
  owner, due, started_at, completed_at, rev, created_at, updated_at; check parent<>id;
depth <= 4 by trigger
plan_edges(from_id, to_id, type blocked_by|relates)  -- trigger rejects cycles
decisions: id, project_id, ref D-0012 (sequential, never reused), title, status
(proposed|accepted|rejected|deprecated|superseded),
  superseded_by, decided_on, decision_makers[], context_md, drivers_md, options jsonb,
outcome_md, consequences_md, confirmation_md, review_by, rev
decision_links(decision_id, target_kind node|doc|decision, target_id, rel
drives|blocks|motivated_by|supersedes)
docs add: owner, review_every_days, last_reviewed_at, reviewed_rev, verified_until
doc_links(from_doc, to_ref, resolved, checked_at)  -- rebuilt on save
```

- Hierarchy: adjacency list + `WITH RECURSIVE` (with CYCLE and a depth limit) [PG1]; a closure table or ltree only if queries get slow [CT1][CT2]; ltree availability on Supabase free UNVERIFIED.
- Ordering: fractional keys as text with COLLATE "C"; keys are case-sensitive and must not be sorted with localeCompare; one owner needs only the unique constraint plus a retry [FI1][FI2].
- RLS: inherit from project_id via the existing private.can_read_project / can_edit_project; wrap auth.uid() in (select …), index project_id and parent_id, name the role in every policy, keep definer helpers in the private

schema with search_path " [S1].

- AI proposals: the AI never writes plan tables. It submits ops {set_status|move|insert|retitle|edit_notes, node_id, base_rev, value} into the Review inbox; accept applies only if rev = base_rev, otherwise the op is stale. AI may only create decisions as `proposed`. Check marks are never auto-applied (a check mark is a claim of completion). Moves are validated for cycles and depth before display.

Progress rules (DESIGN)

1. Only leaf steps count; parents are never stored. Phase % = sum(weight of done leaves) / sum(weight of countable leaves), over all descendants; show n/m beside the %.
2. Cancelled and deferred leaves leave the denominator (Linear lowers scope on cancel [L3]); do not group CANCELLED with done as Obsidian Tasks does [O1].
3. Blocked stays in the denominator as not done; show a red count and the reason from blocked_by.
4. `doing` counts 0 in the headline; optional toggle "in progress at 25%" [L3].
5. Weights default 1; small integers only; show the weight when siblings differ.
6. An empty container shows "no steps", never 0% or 100%; a phase is `done` only when every countable leaf is done; round down so 99.6% never shows 100%.
7. Marking a parent done while children are open: warn and offer "mark all children done"; never cascade silently.
8. Show scope change: store a baseline denominator when a phase starts and show "+N steps added" [B1]; optional uncertain/known flag (cheap hill chart).
9. Compute in a SQL view or RPC, never in a stored column.

Decision record

MADR 4.0.0 (2024-09-17) with optional YAML front matter status/date/decision-makers/consulted/informed [M1]; Y-statement as the one-line summary [Y1]. Lifecycle: proposed > accepted > deprecated | superseded by another record | rejected; never delete or renumber; the old record points to its replacement [NY1]; accepted records are immutable except status and supersession (our rule).

```
---
id: D-0012
status: proposed   # proposed|accepted|rejected|deprecated|superseded by D-00NN
date: 2026-10-21
decision-makers: [owner]
review-by: 2027-01-21
drives: [step:4-3]
---
# Short title naming problem and solution
In the context of <..>, facing <..>, we decided for <..> and neglected <..>, to achieve
<..>, accepting that <..>.
## Context and Problem Statement / Decision Drivers / Considered Options / Decision
Outcome / Consequences / Confirmation / Pros and Cons / More Information
```

Linking: `[[D-0012]]` in a step or document creates a decision_links row on save; a step blocked_by an open/proposed decision shows "waiting on D-0012"; accepting a decision offers to unblock linked steps; superseding shows a badge on old links; review-by feeds the stale list.

Document health, ranked for a solo owner

1. Owner + last_reviewed_at + review_every_days + "verified" state (Notion verified pages: owner, until-date, expiry notice [N2]); an AI-accepted edit does NOT renew the review; reviewed_rev flips the page to "changed since verified" after any later edit; default window about 90 days for fast-changing docs [MB1].
2. Stale/blocked dashboard (overdue review, review window passed, blocked steps, proposed decisions) via pg_cron nightly (Supabase suggests at most 8 concurrent jobs, each under 10 minutes [S2]).
3. In-app link check on save (doc_links; broken links, orphan pages, duplicate titles, steps pointing to a deleted decision); no network needed.
4. markdownlint on save, warnings only (MD001, MD024, MD051, MD052/53); has a browser build [ML1].
5. lychee in CI for external links on exported Markdown, scheduled, not on every push; `--offline`, `--cache`, GITHUB_TOKEN to avoid rate limits [LY1].
6. Vale: CLI only (not browser/Edge); low value for one author [V1]. 7. Drift detection from code: skip.

Templates (DESIGN)

Software: Shape (problem, appetite, rabbit holes, no-gos) > Build > Test > Release > Operate; seed decisions stack/hosting/auth. Research: Question and scope > Sources and method > Collection > Analysis > Writing > Review; step type "claim to verify" with a source link. Course: Outcomes > Syllabus > Content production (module sub-steps, weight by hours) > Practice and assessment > Launch > Revise. Trip: Dates and budget decision > Transport > Stay > Itinerary by day > Bookings (deadlines) > Pre-departure checklist (+ ICS feed). "This master plan": import our `[x]/[~]/[ ]/[?]` Markdown via a status map (~ doing, ? blocked "waiting on owner").

Mistakes to avoid

Stored progress on parents; counting cancelled/deferred/empty phases; blocked treated as done or invisible; AI-accepted edits counted as reviews; parsing the plan only from Markdown text (no stable ids); closure table / ltree too early; integer positions or locale-sorted fractional keys; AI changing checks, decisions or moves without base_rev; cascade delete (use on delete restrict + Trash); editing accepted decisions in place; one global review window; lint that blocks saving; percent alone (also show blocked count, scope added, next step).

Not verified

GFM nested checkbox behaviour; Supabase ltree availability; exact Notion rollup and GitHub sub-issue progress formulas.

Sources

[N1] notion.com/help/tasks-and-dependencies · [N2] notion.com/help/wikis-and-verified-pages · [L1] linear.app/docs/initiatives · [L3] linear.app/docs/project-graph · [G1] docs.github.com/en/issues/tracking-your-work-with-issues/using-issues/adding-sub-issues · [G2] docs.github.com/en/get-started/writing-on-github/working-with-advanced-formatting/about-tasklists · [G3] docs.github.com/en/issues/planning-and-tracking-with-projects/learning-about-projects/about-projects · [O1] publish.obsidian.md/tasks/Getting+Started/Statuses · [O2] blacksmithgu.github.io/obsidian-dataview/annotation/metadata-tasks/ · [OM1] orgmode.org/manual/Checkboxes.html · [B1] basecamp.com/shapeup/3.4-chapter-12 · [B2] basecamp.com/shapeup/1.2-chapter-03 · [M1] adr.github.io/madr/ · [Y1] ozimmer.ch/practices/2020/04/27/ArchitectureDecisionMaking.html · [NY1] cognitect.com/blog/2011/11/15/documenting-architecture-decisions · [PG1] postgresql.org/docs/current/queries-with.html · [CT1] percona.com/?p=2688 · [CT2] dev.to/dowerdev (secondary) · [FI1] github.com/rocicorp/fractional-indexing · [FI2] figma.com/blog/realtime-editing-of-ordered-sequences/ · [S1]

supabase.com/docs/guides/database/postgres/row-level-security · [S2] supabase.com/docs/guides/cron · [MB1] metabase.com/metrics/documentation-freshness (vendor page) · [ML1] github.com/DavidAnson/markdownlint · [LY1] github.com/lycheeverse/lychee · [V1] docs.vale.sh

Research 2026-10-21: MCP security, tokens, Supabase OAuth, AI-edit review, invitee data (feeds phase 6, docs/04)

Tags: VERIFIED = first-party page read; SNIPPET = search excerpt only; UNVERIFIED = not confirmed. Not legal advice.

Bottom line

docs/04 is right in direction but outdated in four places: Supabase OAuth is now verified as a public beta; Client ID Metadata Documents (CIMD) are missing; there is no per-client matrix (most clients can send an Authorization header, so path tokens are only needed for the Gemini app); the Supabase OAuth risk that tokens carry full user privileges is not mentioned. Phase 6 order is wrong: verifying clients (6.6) must come BEFORE OAuth (6.5). The path token is acceptable only as a labelled, short-lived, read-only fallback.

MCP status (VERIFIED, modelcontextprotocol.io/specification/2026-07-28/...)

Latest revision 2026-07-28: stateless (Mcp-Session-Id and initialize removed, `server/discover` added), tasks moved to an extension, multi round-trip `input_required`, `tools/list` results carry `ttlMs` and `cacheScope`, Dynamic Client Registration (DCR) deprecated in favour of CIMD, clients must validate `iss` (RFC 9207), HTTP+SSE deprecated. Authorization: the server is an OAuth 2.1 resource server with Protected Resource Metadata (RFC 9728); clients send RFC 8707 `resource`; servers validate audience and never forward tokens; tokens go in `Authorization: Bearer` and MUST NOT be in the query string; 401 with `WWW-Authenticate` carrying `resource_metadata`. Which clients implement 2026-07-28: UNVERIFIED.

Client	Header token	OAuth	Notes
Claude Code	`--header "Authorization: Bearer …"`, `headersHelper`	`/mcp` login, CIMD, DCR, `--client-id`	VERIFIED code.claude.com/docs/en/mcp. A rejected configured header does not fall back to OAuth. Redirect bug v2.1.229 (127.0.0.1 vs localhost), fixed in v2.1.231. No OAuth in `claude -p`.
claude.ai / Desktop / mobile connectors	"Request headers" beta, limited orgs	CIMD or DCR, redirect https://claude.ai/api/mcp/auth_callback	VERIFIED claude.com/docs/connectors/building/authentication. Uses CIMD only if metadata advertises `client_id_metadata_document_supported: true` and `none` auth; else DCR; needs a real 401 with `resource_metadata`; first `authorization_servers` entry only; 10 s / 30 s timeouts; docs say never accept tokens in the URL.
Cursor	`headers` with `${env:VAR}`	static client id or DCR	VERIFIED cursor.com/docs/context/mcp
VS Code	`headers` with `${input:…}`	`oauth.clientId`	VERIFIED
Gemini CLI	`headers`	401 discovery, DCR, `iss` check	VERIFIED geminicli.com/docs/tools/mcp-server/; page says replaced by "Antigravity CLI" for unpaid/Google One users on 2026-06-18
ChatGPT developer mode	not confirmed	OAuth (CIMD/DCR), none, mixed	SNIPPET; plan eligibility conflicts
Gemini app	UNVERIFIED	UNVERIFIED	Google help (support.google.com/gemini/answer/17209137): add a custom app by MCP server URL on web, US, English, personal account, 18+, Keep Activity on; if no DCR enter credentials manually; writes need confirmation. Plan requirement unclear.

Attacks and mitigations

Official page /specification/2026-07-28/basic/security_best_practices (VERIFIED): confused deputy, token passthrough, SSRF, state hijack, OAuth URL injection, mix-up, localhost redirect impersonation, CIMD trust, scope minimization. We are not a third-party proxy, but with Supabase as authorization server our consent page must name the client and redirect URI, have CSRF protection, anti-framing headers and exact redirect matching; enabling CIMD means we fetch client URLs (SSRF). Tool poisoning/rug pull: practitioner write-ups only (SNIPPET: Willison 2025-07-06, Zuplo, Invariant); mitigation = pin tool definitions by hash on the client. Our real risk is OUR documents as the injection channel (lethal trifecta). Missing in docs/04: exfiltration through rendering — Markdown in proposals can carry remote images/links; if the Review inbox auto-loads them, injected text can leak content before a human accepts.

Tokens

GitHub format (VERIFIED github.blog): type prefix + underscore + 32-bit CRC32 checksum in the last 6 characters, base62; prefix alone cuts false positives to ~0.5%. Partner secret-scanning programme needs a public HTTPS verifier endpoint: too heavy for a solo project; use the format and a gitleaks rule. OWASP: credentials must not be in URLs (logs). SHA-256 of a high-entropy token is adequate (judgement). Path token: Supabase edge logs and Vercel logs contain it (retention unconfirmed) -> only read-only, 24 h to 7 d, low daily limit, one per client, revoked on rotation.

Supabase Auth OAuth 2.1 server (VERIFIED supabase.com/docs/guides/auth/oauth-server/..., changelog 38022)

Public beta (GA target Q4 2025 passed; no GA notice found — check live docs). Available on all plans; signed-in users count to MAU. You supply the consent page ("Authorization Path"). PKCE S256 required; DCR off by default; no CIMD documented. Phase 1 has no scopes: tokens carry full user privileges and rely on RLS, so a leaked MCP token is a normal user JWT that works against PostgREST. `aud` changes only via a custom access token hook; RFC 8707 `resource` not mentioned. Documented defences: RESTRICTIVE RLS on `auth.jwt()->>'client_id' IS NOT NULL`, and an `aud` hook that rejects unknown clients. Access tokens 3600 s, refresh rotates; HS256 default (RS256/ES256 recommended).

Human review of AI edits

Sourced: Google Docs suggestions (colour-tracked, nothing changes until accepted, accept one by one or all, owner emailed — support.google.com/docs/answer/6033474); Claude Code manual vs accept-edits mode (code.claude.com/docs/en/security); Claude connectors tool-level Blocked/Always allow; ChatGPT write actions confirm by default (SNIPPET). Cursor, Copilot, Notion AI, Grammarly, Outline pages could not be fetched (omitted). Over-trust research is thin (SNIPPET arXiv 2412.18531): reviewers dismiss issues assuming the bot would flag them; polished output lowers scepticism; usual mitigation = deterministic gate. Audit trails (SNIPPET, vendor blogs): hash chain per row, optional HMAC, signed checkpoints, triggers blocking UPDATE/DELETE, chain head anchored outside the database; the owner is also DB admin so anchoring matters more than the chain.

Invitee data

Could not fetch GDPR Art. 17 text (one-month deadline is from memory: UNVERIFIED). EDPB summary of an Irish case: emailing non-users without consent/legitimate interest was not justified; "compare and forget" accepted (platform contact imports, not exactly our case). Whether GDPR applies to a solo owner's personal use:

UNVERIFIED. Practical minimum: purge or hash invitee email after accept/expiry; per-person export and delete; log retention period; no tokens or IPs in logs.

Ranked changes

1 (S) run client verification first (Claude Code, Cursor, claude.ai, Gemini app), then make `header` the default token kind and `path` the exception. 2 (S) sanitise the Review inbox: no remote images, link targets as text, diffs as plain text. 3 (S) token format `gfk_…` + CRC32 base62 checksum, show last 4 characters, gitleaks rule in CI. 4 (M) docs/04 items 1-2: two token kinds, rotation with overlap, path tokens 24 h by default. 5 (M) per-token tool and document allow-lists; static tool descriptions; expose a hash of the tool list. 6 (M) OAuth stage: enable Supabase OAuth server with DCR, write consent page (client name, exact redirect URI, CSRF, anti-framing), RESTRICTIVE RLS for `client_id IS NOT NULL`, custom access token hook setting/checking `aud` and rejecting unknown clients, Edge Function returns 401 + `resource_metadata` and validates aud/client_id/iss, keep manual tokens as fallback, test with Claude Code and claude.ai. 7 (M) review friction: reason required for high-risk accepts, no bulk accept for them, auto-accept only a narrow class. 8 (M) invitee data purge/export/delete, log retention. 9 (M) hash-chained audit log + weekly head emailed to an external mailbox. 10 (M) anomaly alerts + auto-pause. 11 (L) support spec 2026-07-28 once the clients you use adopt it. Defer: two-person approval, GitHub partner programme. Risk: building the OAuth stage on a beta feature with full-privilege tokens and no RESTRICTIVE RLS would widen exposure beyond today's read-only path tokens.

Sources (fetched 2026-10-21)

modelcontextprotocol.io/specification/2026-07-28/{basic/authorization, changelog, basic/security_best_practices} · claude.com/docs/connectors/{building/authentication, custom/add-unlisted} · code.claude.com/docs/en/{mcp, security} · cursor.com/docs/context/mcp · code.visualstudio.com/docs/agents/reference/mcp-configuration · geminicli.com/docs/tools/mcp-server/ · supabase.com/docs/guides/auth/oauth-server{,/getting-started,/mcp-authentication,/token-security} · supabase.com/changelog/38022-oauth-2-1-server-capabilities-for-supabase-auth · github.blog/engineering/platform-security/behind-githubs-new-authentication-token-formats/ · docs.github.com/en/code-security/secret-scanning/secret-scanning-partnership-program/secret-scanning-partner-program · cheatsheetseries.owasp.org/cheatsheets/REST_Security_Cheat_Sheet.html · support.google.com/docs/answer/6033474 · SNIPPET: simonwillison.net/2025/Jul/06/supabase-mcp-lethal-trifecta/, zuplo.com/blog/remote-mcp-server-rug-pull/, arxiv.org/pdf/2412.18531, edpb.europa.eu (ie_2022-04), appmaster.io/blog/tamper-evident-audit-trails-postgresql

NOTE 2026-10-21 (owner rule D11, spend nothing): the paid parts of this report (Anthropic API, paid Gemini, any credit or deposit) are DROPPED. Free-only plan: docs/11 phase 7 and prompt Z1 in docs/12. The retrieval design, quote-verification algorithm, evaluation set, injection rules and cost-guard ideas below still apply.

Research 2026-10-21: ask-the-docs (cheapest safe build) — feeds phase 7

(1P) = first-party page fetched. Everything else third-party or UNVERIFIED. The fetch tool summarises pages with a small model, so confirm any number you rely on on the page itself.

Bottom line

Primary: Anthropic API, Claude Haiku 5.5 with the Citations feature, prepaid with a spend cap, about $0.0006 per question (~$0.60 per 1,000). Fallback: Gemini API paid tier (prepay $5 minimum), Flash-Lite, quote-then-verify JSON. Retrieval: Postgres full-text search by headings; add pgvector only if the 40-question test shows weak recall. Do NOT rotate five free OpenRouter accounts (breaks terms, pointless). Do NOT send private docs to a free tier that trains on data (conflicts with D1). NotebookLM (now "Gemini Notebook") cannot be called from the app: personal reader only.

Providers (as of 2026-10-21)

- Anthropic (1P platform.claude.com/docs/en/about-claude/pricing, /api/rate-limits, /build-with-claude/citations): Haiku 5.5 $0.10 in / $0.50 out per MTok up to 100k-token prompts, $0.50/$2.50 above; cache reads $0.01/MTok; Haiku 4.5 $1/$5; Sonnet 5.5 $2/$10; start tier $500/month cap, you can set lower; no free tier; inputs/outputs not used for training by default (privacy.claude.com/en/articles/7996868); citations work on all active models, ZDR-eligible, `cited_text` not counted as output; citations and structured outputs cannot be combined (HTTP 400). Minimum deposit UNVERIFIED.
- Gemini API (1P ai.google.dev/gemini-api/docs/{rate-limits (2026-10-15), pricing (2026-10-19), billing, terms}): free-tier numbers not published (AI Studio shows per project); free tier content may be used to improve products and read by humans; paid not used for training; EEA users require paid services; 3.1 Flash-Lite $0.25/$1.50, 3.5 Flash-Lite $0.30/$2.50 per MTok; prepay minimum $5.
- OpenRouter (1P openrouter.ai/docs/api-reference/limits, /terms): `:free` models 20 req/min and 50/day under $10 of credit, 1,000/day at $10+; extra accounts/keys do not change limits; terms 7(3) bans multiple accounts to circumvent limits; 6.1 some models may train on inputs. -> one account with $10 at most.
- Groq: free-plan figures unclear (gpt-oss-120b 30 req/min, 1K/day, 200K tokens/day, per organization). Cerebras: free trial 5 req/min, 1M tokens/day, "no permanently free tier". Mistral: official limits not found (404). Cloudflare Workers AI: 10,000 neurons/day free, bge-small embeddings $0.02/M tokens (1P, 2026-10-20). Hugging Face: free users no credits. GitHub Models: retired 2026-07-30 (1P). OpenAI: gpt-5-nano $0.05/$0.40 per MTok, text-embedding-3-small $0.02/MTok, no quote API.
- plan file docs/06-API-PLAN.md still says "5 keys rotation": remove it.

Retrieval

A few hundred pages ≈ 0.3-0.6M tokens: do not put everything in the prompt (above 100k tokens Haiku 5.5 costs 5x per token). FTS: stored generated tsvector, GIN index, websearch_to_tsquery, ts_rank in an RPC; heading weight A, body B (1P supabase.com/docs/guides/database/full-text-search). Chunk by h2/h3, keep heading path, split chunks over ~600 tokens at paragraph breaks, store page id and character offsets; top 6 chunks. Embeddings only if recall@6 < ~90%: Supabase hybrid search with RRF (1P supabase.com/docs/guides/ai/hybrid-search), gte-small 384 dims in an Edge Function (docs from 2024, recheck), Gemini Embedding 2 $0.20/M tokens, OpenAI 3-small $0.02/M;

indexing 0.5M tokens costs $0.01-0.10. Gemini File Search tool (managed, 1 GB free store, Markdown supported) ties us to Google with coarse citations: skip.

Verbatim-quote verification (run on every answer)

Option A (primary): send each retrieved chunk as a `document` block with `citations: {enabled: true}`; API returns `cited_text` + location indexes. Option B (any model): JSON `{claims:[{text, chunk_id, quote}]}` or `{refuse:true}`. 1 Normalise quote and chunk: Unicode NFC, collapse whitespace, unify curly/straight quotes and dashes, strip Markdown emphasis markers. 2 Pass only if the normalised quote is a contiguous substring of the named chunk, and the chunk was one of those retrieved. 3 Reject quotes under 6 or over 80 words. 4 Display text from our database using the matched offsets, never the model's copy; link to the heading anchor. 5 If any claim fails retry once naming the failing quote; then show "No verified answer" with retrieved headings as links. 6 Substring proves existence, not support: use the golden set and an optional judge pass.

Evaluation (~$0.03 per run of 40 on Haiku 5.5)

24 single-page facts (8 with numbers/exact strings), 6 two-page questions, 4 paraphrases with little keyword overlap, 6 unanswerable, 2 targeting a planted injection page; hand-written gold page and gold quote each. Metrics: recall@6; quote validity 100%; answer correctness ≥ 90% (owner judged); groundedness (LLM judge "does the quote entail the claim?" + human check of 10); correct refusals 6/6; false refusals ≤ 10%. Re-run after every model, prompt or chunking change; keep history. (RAG Triad per OWASP LLM01:2025.)

Prompt injection

Documents are attacker-controlled (collaborators, AI tools). The Q&A function has no tools (one model call, no web, no writes, no MCP; text only). Never render model output as HTML/Markdown links or images; show plain text and database quotes; link only to our heading anchors. Delimit retrieved text in tagged blocks and say it is data. Retrieve under the caller's identity with RLS. Reuse the phase 6.7 scanner; keep injection items in the golden set. Residual risk: a planted real quote still passes substring: show the source page next to every answer.

Keys and cost guard (1P supabase.com/docs/guides/functions/secrets)

Key only as an Edge Function secret (dashboard or `supabase secrets set`, `Deno.env.get`); names cannot start with `SUPABASE_`; one key for this app in its own provider workspace with its own cap. Never in the bundle (`VITE_`), git, chat, plain DB column, localStorage, MCP token path, logs. Free plan: 500,000 function calls/month, 2 s CPU, 150 s wall clock. Keep JWT verification on; re-check membership in SQL; table `ai_usage(user_id, day, requests, est_cost_micro)` with atomic upsert RPC; ~30 questions/day per user, ~300/day global, kill switch; cap question 500 chars, context ~6k tokens, max_tokens ~600; cache by hash(normalised question, project, content version); hard monthly cap in the provider console (e.g. $5; billing data can lag ~10 min); browser never chooses model or provider.

Gemini app side

NotebookLM / Gemini Notebook: free 50 sources per notebook, each ≤ 500,000 words or 200 MB (1P support.google.com/notebooklm/answer/16215270); compute-based limits (AI Pro 4x, refresh 5 h, weekly cap) (1P support.google.com/gemininotebook/answer/17670842); read-only sharing; no MCP/API mentioned; merge several hundred pages into ≤ 50 files. Gemini app custom MCP app: see docs/14. Gems are removed from personal accounts from November 2026, replaced by skills (1P support.google.com/gemini/answer/15146780): do not build on Gems.

Recommended plan and monthly cost

7.1 answer Q8: no rotation, no free tier for private docs; deposit the minimum on Anthropic, cap $5, own workspace. 7.2 key via Supabase secret dialog. 7.3 FTS chunks + one Edge Function (auth, rate limit, retrieve, Haiku 5.5 with Citations, substring check). 7.4 build the 40-question set first. Later: Gemini paid fallback behind the same function; pgvector only on recall failure. Cost: 1,000 questions ≈ $0.60; 5,000 ≈ $3; total ≈ $1-3/month + one-time ~$5 deposit; with 100k-token contexts ~10x more (the 6k cap prevents it). Risks: a real quote can still mislead or be planted; price jump above 100k-token prompts; free-tier numbers unstable (Gemini, Mistral, Groq UNVERIFIED); free-tier privacy; Gemini custom apps US/English only; EEA needs paid Gemini; model names "Haiku 5.5/Sonnet 5.5" come from the live pricing page, no model was called (latency and citation quality untested).

Research 2026-10-21: Supabase Free / Vercel Hobby / GitHub Actions limits and ops additions (feeds phases 3 and 9)

First-party pages fetched 2026-10-21 unless marked UNV (unverified or third-party). Not checked: the live project's advisors/extension list, third-party keep-alive claims, branching on Free.

Limits

Area	Limit	Source
Supabase projects	2 active; paused ones do not count	supabase.com/pricing
DB	500 MB, shared CPU, 500 MB RAM	pricing
Egress	5 GB + 5 GB cached	pricing
Storage	1 GB total, max file 50 MB (cannot be raised on Free)	/docs/guides/storage/uploads/file-limits
Auth MAU	50,000	pricing
Realtime	200 concurrent connections, 2M messages/month, 100 msg/s, 100 channels per connection, broadcast payload 256 KB, Postgres-change payload 1,024 KB	/docs/guides/realtime/limits
Edge Functions	500k invocations/month; 256 MB; 2 s CPU per request; 150 s wall clock; 100 functions; bundle 20 MB CLI / 5 MB dashboard; outgoing ports 25 and 587 blocked; 100 secrets	/docs/guides/functions/limits
Auth email	built-in SMTP 2/hour and ONLY to project team members; custom SMTP starts at 30/hour	/docs/guides/auth/auth-smtp
Auth rate limits	sign-in 30 per 5 min per IP; refresh 150 per 5 min	/auth/rate-limits
Inactivity pause	after 7 days of "insufficient user database activity"; warning email ~1 week before; restore window 1 year (2024 changelog said 90 days)	/docs/guides/platform/free-project-pausing
Backups	none on Free (export with `supabase db dump`); Pro 7 daily backups; PITR ~100 USD/month, Pro+	/docs/guides/platform/backups
Logs	1 day Free, 7 days Pro	pricing
pg_cron	recommended ≤ 8 concurrent jobs, each ≤ 10 min; Free availability not stated (UNV)	/docs/guides/cron
DB connection	direct connection is IPv6 only; GitHub Actions is IPv4 only -> use the Supavisor session pooler (port 5432); IPv4 add-on Pro only ~4 USD/month	/docs/guides/database/connecting-to-postgres
MFA	TOTP free on all plans; phone MFA paid; leaked-password protection Pro+ (this corrects our earlier note "no MFA on free" — confirm in the dashboard)	search of supabase.com; /auth/password-security
Vercel Hobby functions	300 s max; 2 GB / 1 vCPU; 1M invocations, 4 h active CPU, 360 GB-hrs/month	vercel.com/docs/limits (2026-09-16)
Vercel Hobby transfer/builds	100 GB fast data transfer; 100 deployments/day; 1 concurrent build; runtime logs kept 1 hour	same
Vercel Hobby cron	once per day, up to 59 min jitter	/docs/cron-jobs/usage-and-pricing
Vercel Hobby commercial use	non-commercial personal use only (ads, sales, paid hosting count as commercial)	/docs/limits/fair-use-guidelines

Area	Limit	Source
Vercel Hobby, new 2026-09-16	keeps only the 3 latest production deployments past 30 days; 10 GB deployment storage	vercel.com/changelog/hobby-projects-now-retain-fewer-deployments-to-free-up-storage
GitHub Actions	private repo 2,000 min/month, 500 MB artifacts; public repo standard runners free; scheduled workflows in public repos are disabled after 60 days without repo activity	docs.github.com billing
Pro: Supabase from 25 USD/month (8 GB DB, 250 GB egress, 100 GB storage, 7-day backups, no pausing, leaked-password protection); Vercel Pro 20 USD/month (1 TB transfer, per-minute cron, commercial use).		

What breaks first

1. **Supabase Data API grants change on 2026-10-30** (supabase.com/changelog/45329): new `public` tables are no longer auto-granted to anon/authenticated/service_role; existing tables keep grants; a migration that creates a table without explicit GRANT fails with 42501. Affects `share_links`, `plan_nodes`, `decisions`, `ai_usage` and every future table. (Our 0016 only created functions, which carry explicit grants.)
2. Backups on Free: nothing exists unless we build it; dumps from GitHub Actions must use the session pooler string.
3. Pause risk: our only protection is in-DB pg_cron, which may not count as user activity (undocumented). `/auth/v1/health` does not touch Postgres (UNV).
4. Invitations to real people fail until custom SMTP or an HTTP email API / Gmail function is on (done: /api/invite on Vercel).
5. Realtime is fine for solo + a few people if each tab uses one channel. 6. The 2 s CPU limit hurts heavy work in the MCP function. The 500 MB DB is the likeliest long-term cap (audit logs, versions, embeddings). 7. Hobby is non-commercial. 8. Legacy anon/service_role JWT keys are planned for retirement by end of 2026: migrate to `sb_publishable_` / `sb_secret_` keys (/docs/guides/getting-started/migrating-to-new-api-keys); pg_net and Database Webhooks then need the secret key in the `apikey` header.

Keep the project alive

A real read through PostgREST from OUTSIDE the DB once a day: a daily Vercel cron hitting a function that reads a tiny table, plus a GitHub Actions schedule as second source (note: scheduled workflows in a public repo self-disable after 60 days without activity). Route the pause warning email to a mailbox you read. Pro is the real fix if the project becomes important.

Ranked additions

Phase 3: (S) GRANT in the same migration as RLS for every new table, nothing to `anon` on private tables, matrix assertion `has_table_privilege('anon', t, 'select')` is false; (S) revoke execute on every security definer function from anon/public (lints 0028/0029), definer functions outside exposed schemas with `set search_path = ''`; (S) policies use `(select auth.uid())`, `to authenticated`, index every policy column, no `user_metadata` (lint 0015), `security_invoker = true` on views (0010); (S) run get_advisors after each migration and fail CI on ERROR level; (M) pgTAP (`supabase/tests/database`, `supabase test db`) next to the DO-block matrix (which tests the live project); (S) share_links: hash, constant-time compare, rate-limited lookup RPC, expiry in the RPC, `noindex` + `Referrer-Policy: no-referrer` on that route only; (S) D1 enforced by RLS and RPC, not just the UI. Phase 9: (M) nightly backup via Actions with `supabase db dump --db-url <session-pooler>` (roles, schema, data `--use-copy`), encrypt with `age` (public key in repo, private offline), upload as artifact/release asset not git commits, fail if dump is empty/schema-only; storage bucket separately; (M) weekly restore drill into a throwaway

Postgres service container + row-count checks; (S) keep-alive as above; (S) headers in vercel.json: CSP (`default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'; img-src 'self' data: blob:; connect-src 'self' https://plbvgcnyfclillatpowc.supabase.co wss://plbvgcnyfclillatpowc.supabase.co; font-src 'self'; frame-ancestors 'none'; base-uri 'self'; form-action 'self'` — NOTE our app loads Google Fonts, so font/style hosts must be added; start with Report-Only), Referrer-Policy strict-origin-when-cross-origin (no-referrer on share routes), nosniff, Permissions-Policy camera/microphone/geolocation off, HSTS; `frame-ancestors` only works as an HTTP header; (S) Dependabot (npm + github-actions weekly), `npm audit --omit=dev` + osv-scanner, gitleaks-action; (S) deploy Edge Functions and migrations from CI with a scoped access token (`supabase/setup-cli@v1`, `db push --dry-run` first); (M) axe in CI via @axe-core/playwright with wcag2a/aa and 21a/aa tags; visual regression only with baselines generated on the CI OS/browser; Playwright artifacts: retention 7 days, traces only on failure (500 MB quota); (S) monitoring: free uptime ping of the MCP address; log errors to a small table (Vercel runtime logs last 1 hour on Hobby); (M) search: tsvector generated column + GIN + websearch_to_tsquery; unaccent/pg_trgm availability UNV (check Extensions list); (L) pgvector: gte-small is English-only and truncates at 512 tokens; a few hundred pages need no index or HNSW; 384 floats ≈ 1.5 KB per chunk.

Corrections to our assumptions

Context pack said "no MFA on free": TOTP is free (phone MFA paid). 9.6 "cron stops": the pause criterion is user DB activity; do not rely on in-DB cron as keep-alive. Nightly dump from the direct host fails from Actions (IPv6): use the pooler. Vercel Hobby cron is once a day. Check phase 11 for commercial intent (Hobby is non-commercial). Email from an Edge Function: ports 25/587 blocked; use an HTTPS API (our Gmail path runs on Vercel, not the Edge Function). Phase 3.4 needs grant checks (`has_table_privilege`, `has_function_privilege`) as well as RLS behaviour.

Research 2026-10-21: editor, collaboration, diff, offline/PWA, Markdown, import/export (feeds phases 5 and 8)

Versions/dates from the npm registry fetched 2026-10-21 [N]; sizes from Bundlephobia whole-entry min+gzip [B] (approximate, overstate tree-shaken use). GitHub issue counts mostly UNVERIFIED (API blocked). "mine" = researcher's reasoning.

Editors

Editor	Version (last release)	License	Gzip	Markdown round-trip	Health
CodeMirror 6 (view 6.43.15, merge 6.12.3, lang-markdown 6.5.2)	2026-10-19	MIT	view ~80 KB	lossless (the text is the document)	very active; `@uiw/react-codemirror` 4.25.12 supports react >=17
Tiptap 3 (core/react 3.32.0)	2026-10-13	MIT	react ~9, starter-kit ~106, markdown ~19 KB	`@tiptap/markdown` (MarkedJS) is an "early release", no comments, table cells one child node [tiptap.dev/docs/editor/markdown]	very active
Lexical 0.52.0	2026-09-28	MIT	core ~63 KB	transformer based, no loss statement	pre-1.0
BlockNote 0.56.0	2026-10-14	MPL-2.0	~188 KB	docs call Markdown import "lossy"	active
Milkdown 7.22.3 / Plate (platejs 53.3.18) / MDXEditor 4.3.3	Oct 2026	MIT	not measured	fidelity UNVERIFIED	active
Novel 1.0.2	2025-01-18	Apache-2.0	–	Tiptap based	looks stale
Recommendation: source-faithful Markdown with live preview = **CodeMirror 6** (no round-trip risk; AI diffs, 3-way merges and revisions all work on Markdown text; `@codemirror/merge` gives per-hunk accept/reject from the same dependency; preview via `react-markdown` 10.1.0; inline live preview is later polish). Trade-off: not WYSIWYG. Block editing = **Tiptap 3 + @tiptap/markdown, optional, after CM6**, gated: on open, parse then serialize and compare with the stored text; if different, open source mode only. Avoid BlockNote (lossy, biggest, MPL-2.0), Novel (stale), Lexical (pre-1.0).					

Collaboration

Yjs 13.6.33 (~28 KB), Hocuspocus 4.7.1 (needs a long-lived server; Vercel WebSocket public beta since 2026-06-22, Hobby availability UNVERIFIED), `@supabase-labs/y-supabase` 0.1.0 (pre-1.0; avoid), Liveblocks (hosted; free plan 500 monthly active rooms, 3 members, non-commercial), Automerge 3.5.0 (~1.6 MB gz: cut). Supabase Realtime without CRDT: 200 concurrent connections, 100 msg/s, 20 presence msg/s, 256 KB broadcast (docs/16) — enough for presence. A CRDT is worth it only if 2+ people routinely type in the same document at once with live cursors or long offline merges (mine); a Yjs document is binary, so a Markdown projection must be kept in sync for search, llms.txt and AI edits (and the AI would have to edit through Yjs). **Recommendation 5.2:** `version` column,

save with `update … where version = base`, on conflict 3-way merge with `node-diff3` 3.2.1 (~2 KB), show remaining conflicts as a diff; Presence only for a soft "X is editing" banner; nothing important in Presence.

Diff and suggestions

`@codemirror/merge` `unifiedMergeView({original})` with `acceptChunk`, `rejectChunk`, `getChunks`; `mergeControls` render function since 6.11.0: best fit for per-hunk accept in the Review inbox (baseline = current doc, editor holds AI text). `diff` (jsdiff) 9.0.0 (~8 KB) for headless hunks (`structuredPatch`, apply selected hunks) and revision compare. diff-match-patch stale (2020): skip. prosemirror-changeset only with ProseMirror: skip. Store suggestions as hunks (base version, range, replacement), not editor marks.

Offline / PWA

`vite-plugin-pwa` 2.0.0 (2026-10-03; breaking changes not read), workbox 7.4.1. Quotas per origin, Chrome up to 60% of disk, Safari 17+ ~60% (MDN storage quotas; webkit.org/blog/14403). Eviction removes an origin's data at once; `navigator.storage.persist()` asks for exemption. Safari deletes script-writable storage after 7 days without interaction; WebKit says home-screen apps have their own counter (webkit.org/blog/10218) but developers report contradictions (bugs.webkit.org 237350): test on your iPhone. Android: assume Chromium rules. Private docs (mine): precache app shell only; do not runtime-cache API responses in Cache Storage; keep an allow-listed set (pinned docs) in IndexedDB via Dexie 4.4.6; offline read-only first (offline editing needs an outbox + merge: a second project). Sign-out: delete IndexedDB databases, `caches.keys()`+`caches.delete()`, unregister the service worker, optionally `Clear-Site-Data: "storage"` (MDN: unregisters service workers; `"cache"` behaviour must be tested). Free plan pause means keep your own JSON backup.

Markdown

CommonMark 0.31.2 (2024-01-28); GFM spec 0.29 (2019), looks stale. Front matter is not standard: `remark-frontmatter` 5.0.0 + `yaml` 2.9.1. unified 11.0.5, remark-parse 11.0.0, remark-gfm 4.0.1, rehype-sanitize 6.0.0, react-markdown 10.1.0: mature, slow-moving. Math: remark-math 6.0.0 + rehype-katex 7.0.1 + katex 0.19.0 (~77 KB, lazy). Mermaid 12.2.0 (~174 KB, lazy-load when a document has a mermaid fence; default `securityLevel: 'strict'`; `sandbox` beta). Sanitization: rehype-sanitize default schema follows GitHub's; place it after the last unsafe step; it prefixes ids/names with `user-content-` (breaks in-page anchors: our "stable block anchors" need a fix-up); for math sanitize first with a schema allowing `math-inline`/`math-display`, then katex; never `allowDangerousHtml` without sanitizing. `[[kind:slug]]` needs a small micromark/remark plugin.

Import / export

`fflate` 0.8.3 (~12 KB) preferred over `jszip` 3.10.2 (~28 KB; CVE-2022-48285 path traversal fixed in 3.8.0). Untrusted zip checklist (SonarSource RSPEC-5042; implementation ours): reject `..`, absolute paths, backslashes, drive letters, NUL; cap entry count, total and per-entry uncompressed size; check compression ratio (legit ≈ 1-3); enforce limits while decompressing; do not trust extensions; run in a Web Worker with a timeout; fflate's built-in limits unknown, add ours. `.docx`: `mammoth` 1.13.0 performs no sanitisation, can emit `javascript:` links, crafted files can burn CPU/memory, Markdown output deprecated: docx -> HTML (mammoth) -> sanitize -> Markdown (`turndown` 7.2.4). Notion export: zip of CSV + Markdown, callouts exported as HTML (notion.com/help/export-your-content); third-party (updated Feb 2026): 32-hex ID suffix per file, images in subfolders, internal links reportedly not preserved (UNVERIFIED): write a resolver. Obsidian: `[[Note]]`, `[[Note|alias]]`, `[[Note#Heading]]`, `[[Note#^block]]`, `![[embed]]`; on export use standard Markdown links with URL-encoded paths (obsidian.md/help/links).

Stack for phases 5 and 8

1 CodeMirror 6 + @codemirror/merge + react-markdown preview. 2 version-column save + node-diff3 + Presence banner. 3 jsdiff for revision compare/headless hunks. 4 fflate + mammoth + turndown in a Worker with sanitizing and caps. 5 vite-plugin-pwa precaching the shell only + IndexedDB pin list. Cut/defer: CRDT/Hocuspocus/Liveblocks/Automerge, Tiptap block editor (later, gated), offline editing and background sync, inline live-preview decorations, semantic search, Mermaid sandbox, Notion link-graph reconstruction, rich suggestion marks, PDF export (use print CSS). Mistakes to avoid: WYSIWYG that rewrites Markdown on save; caching authenticated API responses in the service worker; not wiping caches/IndexedDB at sign-out; sanitize order mistakes (rehype-raw, KaTeX); unzipping without caps; mammoth Markdown output; relying on y-supabase or Novel; trusting a Safari quota number; assuming a home-screen app is safe from eviction. Not checked: Milkdown/Plate/MDXEditor round-trip; Safari 7-day rule and Clear-Site-Data on real devices; Vercel Hobby WebSockets; jsdiff v9 release notes. Sources: registry.npmjs.org/; bundlephobia.com/api/size; codemirror.net/docs/changelog/ and /docs/ref/#merge; tiptap.dev/docs/editor/markdown; lexical.dev/docs/packages/lexical-markdown; blocknotejs.org/docs/features/import/markdown; github.com/{Milkdown/milkdown, steven-tey/novel, mdx-editor/editor, supabase-community/y-supabase}; liveblocks.io/docs/pricing/plans/free; vercel.com/changelog/websocket-support-is-now-in-public-beta; developer.mozilla.org Storage_quotas_and_eviction_criteria, Clear-Site-Data; webkit.org/blog/{10218,14403}; spec.commonmark.org; mermaid.js.org/config/usage.html; github.com/rehypejs/rehype-sanitize; github.com/mwilliamson/mammoth.js; notion.com/help/export-your-content; obsidian.md/help/links; vulert.com/vuln-db/CVE-2022-48285; rules.sonarsource.com RSPEC-5042.

Research 2026-10-21: document-workspace benchmark vs our master plan

Legend: Y yes, P partial, N no/not found, ? UNVERIFIED (no first-party page reached), $ paid, EE enterprise. A "?" is a gap in evidence, not proof of absence. Universal basics (tree, move, trash, Markdown export) omitted.

Capability matrix (products: Notion, Outline, Docmost, AFFiNE, AppFlowy, BookStack, Confluence, GitBook, Slab, Obsidian, Anytype)

- Verified/owner/expiry: Notion Y$, Outline N, Docmost EE, Confluence P, GitBook P, Slab Y on the free plan, Obsidian N. (N2 notion.com/help/wikis-and-verified-pages; S1 slab.com/pricing; S2 help.slab.com/en/articles/4136379-post-verification)
- Per-page permission override: Notion Y, Outline Y, Docmost Y$ (Business), AppFlowy Y (v0.9.4), BookStack Y, GitBook roles only.
- Guests: Notion Y (10 free), Outline $, Slab Y (3/user paid), Anytype Y. Comments + mentions: Outline Y (read-only users can comment), Docmost P (resolve EE), GitBook Y (change-request review).
- Version history: Notion 7 days free, Docmost free + compare (v0.96), Slab 90 days free, Obsidian 1 month/12 months paid. Public share: Notion Y (expiry), Outline Y, Docmost Y free, GitBook Y (private links paid), Obsidian Publish.
- API/webhooks: Notion, Outline, BookStack (webhooks since v21.12), Confluence, GitBook, Slab Y; Docmost API EE. MCP: Notion, Outline, GitBook Y; Docmost EE.
- Audit log: Enterprise in Notion, Slab, Docmost, GitBook; BookStack Y (admin). Page analytics: Notion basic free, GitBook all plans, Slab 30 days free, Confluence Premium+ (1-year lookback, privacy mode).
- Notifications inbox: Notion, Docmost (v0.71), GitBook, Slab. Favorites/recents: Notion, Docmost (v0.80). Search: Notion split keyword vs AI on 2026-09-02; Docmost filters v0.96, vector EE; Slab facets (verified/expired).
- AI Q&A: Notion $20 Business, Outline $, Docmost EE, AppFlowy $8.9 add-on, GitBook Essential+, Slab Business+. SSO/SCIM mostly paid tiers. Sources: notion.com/{pricing,help/sharing-and-permissions,help/use-pages-offline}, developers.notion.com/page/changelog, github.com/outline/outline/releases, getoutline.com/{changelog,pricing}, releasebot.io/updates/docmost, docmost.com/{docs,pricing}, affine.pro/pricing, appflowy.com/pricing, bookstackapp.com/blog/, support.atlassian.com/confluence-cloud/docs/view-analytics-to-see-how-content-is-performing/, gitbook.com/{docs/changelog,pricing}, slab.com/pricing/, obsidian.md/{pricing,help/Obsidian+Sync/Version+history}.

15 highest-value gaps for a solo owner on free tiers (with recommendation)

1 **Verified / owner / review-by** (Slab on free plan: primary verifier, review frequency, green banner with expiry, yellow once lapsed, diff since last verification; Notion: owner required, expiry date or indefinite, inbox + email at expiry, search badge). Add owner_id, verified_by, verified_at, verified_until; daily pg_cron flips to expired and inserts a notification; badge, filter, "changed since verified" diff. 2 **Trust-weighted search and ask-the-docs** (Notion ranks verified pages higher, notion.com/releases/2025-03-26): rank boost in FTS; mark each cited quote verified/unverified; allow "verified documents only". 3 **Real notification inbox** (2.29): one `notifications` table (AI edit waiting, verification due, mention, role change) over Realtime; skip the email digest. 4 **Request access + a `restricted` flag** instead of full per-page ACLs (Outline added "Request access" 2026-06-01): keep org/project

roles; one boolean per document; button creates a notification. 5 **Share-link hardening**: the `#/s/` payload must contain no editor ids or emails; last_viewed and view count per link; extend the anon SQL matrix to the share route. 6 **Page view analytics**: `doc_view_daily(doc_id, day, count)` upsert RPC, no per-view rows; 30-day views, last viewed, "unread 90 days" as staleness signal. 7 **Revision attribution, retention, provenance** (Notion 2026-10-13, Outline v1.10/v1.11 reasons): `revisions.actor_kind` (human | ai-token | import) with the token name; keep last N plus every verified; optional reason on delete. 8 **Per-token usage panel** + `report_issue` **MCP tool** (GitBook Agents & MCP report 2026-09-22; sendFeedback 2026-07-29; Notion admin can see last used per AI tool): last used, call count, documents touched per token from the audit log; `report_issue(doc, text)` lands in the Review inbox; defer OAuth. 9 **Content-gap list** (GitBook, 2026-10-06): log every question with no verbatim quote found, with a count; top ten on the dashboard. 10 **Templates, small**: `is_template` flag, "New from template" copies the page, no variables yet. 11 **Favorites** table (user, doc); recents stay in localStorage (as the palette). 12 **Offline: read cache only** (Notion offline is per-device, no subpage/permission sync): service worker + IndexedDB of last N opened documents. 13 **Human suggestions and comment resolution**: viewers' "suggest" edits go through the Review inbox; comments carry an anchor quote and a resolved flag, no threads yet. 14 **Minimal import/export with an exit guarantee**: Markdown zip in and out + Notion export zip import; DOCX later; keep zip export working always. 15 **Search filter chips** (project, kind, verified state, owner, labels); keyword-only until the 40-question set fails.

Over-engineered for a solo owner (cut or defer)

6.7 two-person approval; 6.4 hash chain and replay (plain append-only log + CSV is enough; audit logs are Enterprise tier elsewhere); 6.3 rotation with overlap / new-client alert (keep kill switch and per-token limits); 6.5 OAuth server (Docmost makes MCP OAuth Enterprise-only; tokens suffice for one owner — but see docs/14 for client compatibility); 8.4 pgvector until FTS fails the evaluation set; 4.6 timeline/board/ICS (list view only); 5.9 attachments (Notion free caps 5 MB, Slab 10 MB; use the 1 GB bucket, images only); 2.23/Q11 move project across organizations (export/import covers it); 2.25 chords, 2.27 visual-regression screenshots, 2.28 logos/accents; 9.10 domain + Resend until you invite more than a few people; 8.5 listen/focus/split view; 5.8 four importers at once (start with Markdown zip); 9.3 backup encryption until backups leave a private repo.

Mistakes and risks other products hit

- Public pages leaking editor identity: April 2026 reports say public Notion pages expose editors' names/emails via an unauthenticated API (cyberinsider.com/notion-pages-have-leaked-user-data-via-an-unauthenticated-api-since-2022/; secondary, fix status UNVERIFIED).
- AI edits taking effect before approval (PromptArmor: promptarmor.com/resources/notion-ai-unpatched-data-exfiltration; simonwillison.net/2025/Sep/19/notion-lethal-trifecta): our Review inbox is the right design; a pending diff must never auto-load remote images or links.
- BookStack 2026 security releases are almost all permission leaks (drafts, attachments, comments on recycled pages via API, hidden content in Markdown exports, search showing drafts): extend the RLS matrix to drafts, trash, attachments, exports, search, comments.
- Atlassian April 2022 outage (a script with wrong ids deleted ~400 customers' sites; fix = universal soft delete; atlassian.com/blog/how-we-build/post-incident-review-april-2022-outage): keep soft delete and add a TESTED restore from backup.
- Obsidian Sync silent overwrites (forum.obsidian.md/t/obsidian-sync-on-iphone-overwrites-newer-data-causing-data-loss/85214; anecdotes): keep both versions on conflict; no offline writes in v1.
- Confluence "Verified" cleared by any edit, no timestamp (CONFCLOUD-84852): store verified_at and decide explicitly what an edit does. Rubber-stamp risk: do not accept a bare "still accurate" button (Slab shows a diff since last verification).

- Vendor comparison pages can be wrong (one said Slab has no verification; Slab's own pricing lists it on free): trust first-party pages.
- Notion API churn (2026-03-11 `archived` -> `in_trash`; 2026-07-15 page links moved to app.notion.com; per-workspace rate limit 2026-06-16; notion-search keyword-only 2026-09-02): version our MCP tool schemas and key `[[kind:slug]]` links on ids. AI pricing shifts (AI bundled into Business at $20/member): keep AI off the free path, bring-your-own key.
- Supabase free pause: see docs/16. Not checked: Outline mobile, AFFiNE and Anytype 2026 changelogs, Docmost/AppFlowy permission details beyond citations, Confluence native expiry.

Audit of the ChatGPT Deep Research report, Parts A + B (2026-10-22)

Raw report: `19a-RESEARCH-AB-CHATGPT-RAW.md` (14.6k words). Audited by Claude against registries, Supabase docs and the live project. Verdict: strong report, usable, but it did not see our code or live schema and has the errors below.

Verified TRUE (primary source)

- Supabase Data API: opt-out setting 2026-04-28; default for new projects 2026-05-30; all existing projects 2026-10-30. Existing objects keep grants. Official opt-in SQL covers tables, sequences AND functions (docs "Securing your API"): `alter default privileges for role postgres in schema public revoke select,insert,update,delete on tables from anon,authenticated,service_role;` + `revoke execute on functions from anon,authenticated,service_role;` + `revoke usage,select on sequences ...;` + `revoke execute on functions from public;`. Dashboard setting: Integrations > Data API > "Default privileges for new entities". Docs also recommend a dedicated `api` schema as an extra boundary.
- React 19.3 (2026-09-09): View Transitions + Fragment refs stable; 19.3.1 on 2026-10-14. Vite 8 uses Rolldown as the single bundler, Node 20.19+/22.12+. MADR 4.0.0 = 2024-09-17. Anchor positioning: webstatus "limited" (Safari 27 available 2026-09-14), keep as enhancement only.
- GitHub Actions: standard runners free for public repos; private Free plan 2,000 min/month, 500 MB artifacts, 10 GB cache.

WRONG in the report

- "Supabase Free has managed DB backups": FALSE. Supabase backups doc: daily backups only for Pro/Team/Enterprise; Free should export its own data (`db dump`). Master plan was right.
- Version ledger: merge is 6.12.3 (report 6.12.2, called the PDF stale: the PDF was right); react 19.3.1 (not 19.3.0); zustand 5.0.16 (published 2026-10-22); react-aria-components 1.22.1.
- Missed: functions default privileges (see above). Missed our repo state: already Vite 8.3.4, TS 7.0.2, plugin-react 6, React 18.3.1; browser/invite/MCP already use publishable/secret keys (MCP reads SUPABASE_SECRET_KEYS, falls back to SERVICE_ROLE var).
- SQL pitfalls in its samples: pgcrypto lives in schema `extensions`, so `digest()` must be `extensions.digest()` under `search_path=''`; unique INDEX on (project_id,parent_id,order_key) cannot be deferred so a reorder/renumber UPDATE can hit transient duplicate errors (needs deferrable constraint or two-step renumber; not tested); sample schemas assume names that differ from ours (ours: docs.body not body_md, doc_revisions.rev not revision_no, docs.position integer, docs.parent_ref, private.can_read_project/can_edit_project already exist).

Live facts (checked 2026-10-22)

PG 17.11; DB size 15 MB of 500 MB (largest table 296 kB); unaccent available; 33 public tables all RLS on; 42 definer functions in public, 28 executable by authenticated (lint 0029), 0 by anon; 35 functions in private schema. anon has SELECT grant but no anon policy on workbench_state, app_admins, project_members (harmless, revoke for least privilege). Public anon reads (docs/doc_revisions/projects) go through private.can_read_project (built-in project is public by design).

Owner decisions raised

1. Canvas single-file build uses React 18; upgrading to 19 may break Canvas. Keep Canvas or drop it?
2. Report says use native `<select>` for simple lists; owner asked for custom animated dropdowns. Keep custom?
3. Share link: document-only first (report recommends) or project-wide?

Remaining research (see chat): functions default-privilege behaviour on 10-30 for new RPCs; Realtime 2026 pattern (broadcast-from-DB vs postgres_changes); API-schema/RPC-only architecture; revision retention (low priority).

Audit of the ChatGPT Deep Research report, Part C: security and privacy (2026-10-22)

Raw report: `21a-RESEARCH-C-CHATGPT-RAW.md` (15k words; its 70-item security test pack and 37-item malicious-input corpus M01-M37 live there and are the test source). Audited by Claude against primary docs, npm, the live site, the live database and the full git history. Verdict: good report; its design is sound; several of its "2026 facts" were wrong or unconfirmed.

Verified TRUE

- Supabase Auth (docs): TOTP app-authenticator MFA exists; `aal` claim in the JWT; RLS pattern `(select auth.jwt()->>'aal') = 'aal2'` (docs recommend `as restrictive` policies). Sessions last indefinitely by default with unlimited sessions; time-box, inactivity timeout and single-session are **Pro-only** (docs). Refresh-token reuse detection exists (10 s reuse interval). Passkeys are **experimental** (needs supabase-js >= 2.105.0). CAPTCHA supports hCaptcha and Cloudflare Turnstile. The default SMTP is best-effort, non-production and rate limited ("can change without notice"). Supabase account MFA: no recovery codes, register a backup TOTP factor (that page is about dashboard accounts; project-user recovery codes still UNVERIFIED).
- Incidents (searches, several sources): Axios npm compromise 2026-03-31 (hijacked maintainer, versions 1.14.1 and 0.30.4, fake dependency `plain-crypto-js@4.2.1` with a postinstall RAT, live about 3 h); Shai-Hulud npm worm 2025-09 and Shai-Hulud 2.0 2025-11; GitHub's npm hardening plan after the first wave. Mammoth CVE-2025-11849 (GHSA-rmjr-87wv-gf87, published 2025-10-17, critical 9.3, fixed in 1.11.0). Confluence CVE-2023-22515/22518 are real.
- npm `min-release-age` (days) exists since npm 11.10.0 (two secondary sources). Push protection and secret scanning are free on public repos (GitHub blog); no historical scan in the free push protection.
- Fragment principle: a URL fragment is not sent in the HTTP request and is excluded from Referer (MDN, Referrer Policy spec); it does NOT protect against history, extensions, screenshots or JS reading `location`.

WRONG or UNVERIFIED in the report (do not rely)

- WRONG (not confirmed, contradicts docs): "Vercel made production deployment protection free on all plans in September 2026". Vercel's docs say Vercel Authentication covers preview/deployment URLs on Hobby; protecting the production domain needs Pro. Do not plan on it.
- UNVERIFIED: Mammoth prototype-pollution (fixed 1.12.2) and ReDoS (fixed 1.12.3) advisories (not found); npm shows mammoth 1.12.1-1.12.3 and 1.13.0 (2026-09-26) exist. Policy: pin the latest release, recheck advisories before install, sandbox anyway. `allow-scripts` npm option: not found. Microsoft Safe Links behaviour on fragment pages. Gmail app-password daily limit (the old "500/day" is unverified; never rely on it). Notion/Google Docs/BookStack/GitHub leak case studies (no primary source).
- Over-specific: HSTS. Vercel ALREADY sends `strict-transport-security: max-age=63072000; includeSubDomains; preload` on the live site: do not set HSTS in vercel.json. Live site also sends `access-control-allow-origin: *` on static files; `/share` returns 404 today; index.html contains a `<style>` tag.
- CodeQL on public repos: the GitHub page only says private repos need a Code Security license; public availability is implied, not stated: check Settings > Code security.

Live facts (checked 2026-10-22)

- Full git history (53 commits, all authored "Claude (noreply@anthropic.com)"): no secrets (sb_secret, JWTs, Google/GitHub tokens, app passwords, private keys, DB URLs), no real email addresses (only `*@example.test` and one `onboarding@resend.dev`), no zip/env/trace files ever tracked. The repo can stay public from a data-leak view; run gitleaks in CI anyway.
- Today: no `vercel.json`, no CSP, `/share` not routed, CI publishable key literal in ci.yml (public key, fine), no scanners, no MFA, password-only sign-in, no in-app account creation.

Design adopted into the master plan (audit recommendations, owner OK pending where marked Q)

1. Share link = `https://<app>/share#<43-char base64url secret>` (not `/#/s/...`): Vercel rewrite `/share` to index.html so headers/robots can differ; first paint shows nothing; explicit "Open shared project" click; POST the secret in the body; `history.replaceState` to `/share`; content only in memory; no analytics/third parties/service-worker caching on that page.
2. Redemption through a SERVER proxy (Vercel function with `sb_secret_` in a server env var, SQL function executable only by service_role). Reject: anonymous SECURITY DEFINER RPC (needs anon API execute) and share-viewer JWTs. `Cache-Control: no-store`; revocation checked on every fetch; global kill switch; pseudonymous short-lived rate bucket, no raw IP log.
3. Several independent project-wide links (labelled, read-only, default 7 days, no never-expire). Store only SHA-256 of 32 random bytes; slow hashes and homemade constant-time compare are not needed (indexed digest lookup). The SQL must use `extensions.digest` under an empty search_path.
4. Public projection allowlist: project name/description; documents (opaque key, title, current body, order); plan (title, hierarchy, status). Everything else excluded (members, emails, ids, revisions, review inbox, drafts, Trash, comments, tokens, audit, notifications, search tables, Storage, Realtime, exports). No Storage images in v1 shares.
5. Auth: public sign-up OFF; invite-bound account creation through `/api/join` using `auth.admin.createUser(email_confirm:true)` with a server key; existing accounts must sign in first; Turnstile on that endpoint; generic errors; password minimum 15 (NIST 800-63B-4, single factor); TOTP for owner/admin; AAL2 required inside sensitive RPCs; magic links not the primary flow; passkeys later.
6. Browser hardening: move the inline CSS string and the `<style>` in index.html to a real CSS file; drop Google Fonts (system stack or self-hosted); `vercel.json` with CSP in Report-Only first (default-src 'none'; script-src 'self'; style-src 'self'; connect-src self + Supabase https/wss; img-src self blob: Supabase; frame-ancestors 'none'; base-uri/object-src none), global Referrer-Policy no-referrer, nosniff, X-Frame-Options, Permissions-Policy, COOP/CORP same-origin, global X-Robots-Tag noindex, `/share` no-store, hashed assets immutable. NOT HSTS. Skip a public CSP-report endpoint in v1 (extra anonymous surface): catch violations in Playwright. zip.js workers may need `worker-src 'self' blob:`.
7. Untrusted content: no raw HTML, no remote images in review/share, URL allowlist https/mailto/#/internal, `user-content-` id prefix, no SVG upload, YAML front matter allowlist (no custom tags/aliases, reject `__proto__`/`constructor`), CSV neutralisation, ZIP import first (`@zip.js/zip.js` 2.24.0 or fflate 0.8.3 pinned) in a Worker with caps (25 MiB archive, 100 MiB emitted, 10 MiB entry, 1000 entries, ratio 100:1, depth 20, reject symlinks/absolute/`..`/NUL/reserved names/case collisions); `.docx`, Notion zip deferred (Mammoth latest + Worker + external access disabled).
8. Repo/CI: Actions pinned by full SHA, `permissions: contents: read`, no `pull_request_target` with untrusted code, gitleaks (+ custom `sb_secret_` rule), osv-scanner, `npm ci`, `npm audit --omit=dev`, Dependabot with cooldown, npm `min-release-age=7`, try `npm ci --ignore-scripts`, no server secrets in

the build environment, GitHub secret scanning + push protection + Dependabot + code scanning where available, ruleset requiring CI, private vulnerability reporting.

9. Privacy: delete the duplicate invite email after the invite is resolved (hashing is not deletion); pseudonymise authors on user deletion; invitation preview reveals nothing to unauthenticated visitors; bounded retention for security events.
10. Audit: small append-only `private.security_audit_log` (reject UPDATE/DELETE by trigger, no grants), security events only; no hash chain now; pgaudit only for investigations; Vercel Hobby runtime logs are about 1 hour.
11. Abuse/invites: `/api/invite` requires auth + AAL2 + an existing invite row, recipient and body derived server-side; quotas 3/hour and 10/day per org, 3/day per recipient (policy, replaces 20/60); owner kill switch table `private.security_settings`.
12. Runbook for the sole owner (credential inventory, share-link kill, secret rotation, account recovery, compromised GitHub/Vercel/Supabase/Google) before public sharing ships.

Open questions for the owner

Q22 project-wide share vs per-document `restricted` flag (report: they conflict); Q23 TOTP mandatory for whom; Q24 keep the repo public (history is clean).

Audit of the ChatGPT Part E+F report (operations + email), 2026-10-22

Raw report: `docs/22a-RESEARCH-EF-CHATGPT-RAW.md` (about 2,250 lines; its code is marked "untested" by its author). Every fact below was re-checked on 2026-10-22 against the primary page named in brackets. UNVERIFIED = I could not reach a primary source; do not build on it without a test.

1. Verified TRUE (safe to plan on)

Claim	Evidence
Supabase pauses Free projects with low activity over 7 days; "a few user requests to the database each day" is typically enough; warning email about 1 week before, confirmation after	[supabase.com/docs/guides/platform/free-project-pausing]
Restore window after a pause is 1 YEAR (body text of the live page). NOTE: the Supabase MCP `search_docs` index still returned "90 days", so that index is stale; the live page wins. docs/16 already had this right	live page fetched; MCP index conflicting
Free plan has no downloadable backups; production checklist says the same	[going-into-prod]
Edge Functions: "Outgoing connections to ports 25 and 587 are not allowed" (465 is simply not listed = not documented as blocked, not guaranteed). Free wall-clock 150 s, 256 MB memory, 2 s CPU per request	[functions/limits]
Vercel Hobby cron: once per day maximum, timing "per-hour (±59 min)", 100 cron jobs per project	[vercel.com/docs/cron-jobs/usage-and-pricing, updated 2026-07-15]
GitHub: scheduled workflows run only on the default branch, can be delayed under load, minimum 5 min, and are disabled in a PUBLIC repo after 60 days without repository activity	[docs.github.com events-that-trigger-workflows]
Artifacts and logs are kept 90 days by default; public repos can set 1 to 90 days	[docs.github.com retention page, via search]
Gmail app passwords: need 2-Step Verification; **unavailable if 2SV is set up only with security keys, for work/school accounts, or with Advanced Protection**; revoked when the Google password changes	[support.google.com/accounts/answer/185833]
Healthchecks.io Hobbyist (free): 20 checks, 100 log entries per check; no multiple accounts to dodge limits	[healthchecks.io/pricing]
UptimeRobot Free: 50 monitors, 5-minute interval, 1 basic status page, 5 integrations, "good for hobby and non-profit projects", "no credit card required"	[uptimerobot.com/pricing]
Neon Free: 100 projects, 1 GB per project (20 GB account), 100 CU-hours per project, scale to zero after 5 min, 6 h restore window (1 GB), no card	[neon.com/pricing]
Backblaze B2: "First 10GB storage is always free"; Class A/B/C calls free, Class D first 2,500/day free; egress free up to 3x monthly average storage	[backblaze.com/cloud-storage/pricing]
`supabase db dump` excludes Supabase-managed schemas ("auth, storage, and those created by extensions"); the official restore recipe uses three dumps (roles `--role-only`, schema, data `--use-copy --data-only`) and `psql --single-transaction -v ON_ERROR_STOP=1 ... -c 'SET session_replication_role = replica'` between schema and data	[CLI reference supabase-db-dump; self-hosting/restore-from-platform]
Direct DB connections need IPv6 (or the paid IPv4 add-on); the shared pooler (Supavisor) is IPv4. GitHub-hosted runners therefore need the **session pooler** URL	[pooling-and-limits]
With custom SMTP the default Auth email rate limit is 30 new users per hour (dashboard-configurable); default Supabase SMTP is best-effort, no SLA	[going-into-prod, auth-smtp]

2. UNVERIFIED or only third-party (keep marked)

- B2 sign-up "no billing method needed": the pricing page is silent; a web search summary of Backblaze's own sign-up page says "no credit card needed" and several third-party guides say a **phone number must be verified**. The sign-up page itself returned 403. Treat as: probably no card, probably a phone number. Owner must try it (see questions).

- Personal Gmail daily limit (about 500/day): the Google page I could fetch covers Workspace only (2,000/day per user, 24 h block when exceeded). Our own cap is 10/day per organization, so the exact Gmail figure does not matter.
- `smtp.gmail.com:465` as host/port: no current first-party page retrieved by the report or by me. Prove with one real send before relying on it. Same for entering it in Supabase Auth custom SMTP.
- Whether a PostgREST read counts as "user database activity": best-supported, not contractually defined. `pg_cron`, Auth-only calls and no-DB Edge calls: do not rely on them.
- Exactly which auth/storage/vault objects `supabase db dump` writes: the CLI reference says schema dump skips auth and storage, while the self-hosting guide says the data dump contains `auth.users` and COPY statements for auth/storage tables. These two statements conflict, so inspect the first real dump (test 14 below) before designing the restore.
- Vercel Hobby deployment-retention change of 2026-09-16 (report cites only a search snippet): not verified; irrelevant because we never use deployment history as an archive.
- Scanner behaviour (Proofpoint, Mimecast, Slack previews: GET vs HEAD, JS execution): UNVERIFIED. Our design (secret in the fragment, explicit click, POST) does not depend on it.
- ChatGPT R2 "needs checkout": report says unproven either way; R2 stays rejected for the no-card rule.

3. Defects in the report's sample code (do NOT copy as is)

1. `backup.yml` weekly drill condition `github.event.schedule == '23 1 * * 0'` can never be true: only one cron (`23 1 * * *`) is declared, so the "weekly" restore test would run only on manual dispatch. Fix: one daily cron and decide inside the job (`date -u +%u` = 7).
2. The restore test loads `schema.sql` and `data.sql` into a plain `postgres:17` service and skips `roles.sql`. Our schema references roles `anon`, `authenticated`, `service_role`, the `auth` schema (`auth.uid()`, FKs to `auth.users`) and extensions (pgcrypto in schema `extensions`, supabase_vault, pg_cron). On vanilla Postgres this fails at the first policy. The test must restore into a Supabase-shaped target: either `supabase start` (local stack, accurate, a few minutes) or a vanilla container plus a small prelude that creates those roles, schemas and `auth.users`. Pick after inspecting the first real dump.
3. `npm install --global supabase` is not the documented route (docs describe a per-project dev dependency, version pinned). Use the official `supabase/setup-cli` action (already in plan item 9.6), pinned by commit SHA, with a pinned CLI version.
4. `actions/checkout@v4`, `actions/upload-artifact@v4` use floating tags and checkout is not needed at all. Our rule (plan 9.5): pin every third-party action by full commit SHA; drop checkout.
5. `curl https://rclone.org/install.sh | sudo bash` is exactly the supply-chain pattern we ban. Use the distro package (`apt-get install rclone`) or a pinned release zip with a checked SHA-256.
6. `SUPABASE_DB_URL` must be the **session pooler** string (user `postgres.<project-ref>`, port 5432, host copied from the Connect dialog), not the direct host (IPv6 only). The report says this in prose but the workflow gives no hint.
7. Env names in `/api/ops-keepalive.ts` (`SUPABASE_URL`, `SUPABASE_PUBLISHABLE_KEY`) differ from the ones already used by `api/invite.ts` (`VITE_SUPABASE_URL || SUPABASE_URL`, `VITE_SUPABASE_PUBLISHABLE_KEY || SUPABASE_PUBLISHABLE_KEY`). Reuse the existing lookups; no new variables on Vercel.
8. Healthchecks pings only on success. Add a `/fail` ping on failure (faster alert than waiting for the grace period) and a `/start` ping for duration.
9. The artifact upload happens before the B2 upload and the job has no `environment:`; put the secrets in a GitHub Environment limited to `main` so only `main` workflow runs can read the DB password and B2 key.

10. `ops_keepalive` migration is fine (explicit GRANT, which the 2026-10-30 change requires). Add: the table gets no sequence; keep it out of the Realtime publication; add it to the grants test.
11. Dump role: the report uses the `postgres` password. A narrower role would be better but dumping RLS-protected tables needs a role that bypasses RLS (otherwise rows are silently omitted or the dump errors). Whether the managed `postgres` role can create such a role is UNVERIFIED; v1 uses `postgres` with the password stored only as an Environment secret; revisit.

4. Adopted design (what goes into the master plan)

1. **Keep-alive (9.2):** table `public.ops_keepalive` (one row, public read, no writes), read by (a) a Vercel cron `/api/ops-keepalive` once daily and (b) a GitHub Actions schedule at a different hour. Both ping Healthchecks on success and `/fail` on error. Vercel's timing varies by up to 59 minutes, so the dead-man grace is at least 2 h. The GitHub schedule disables itself after 60 days without repo activity; the Vercel cron is the independent second source. The Supabase pause-warning email goes to a mailbox the owner reads.
2. **Monitoring (9.8):** UptimeRobot (HTTP check of the site and of a `/api/health` endpoint that really reads one row; 5 min) + Healthchecks (keep-alive x2, backup, storage export, restore drill). Two free accounts, one purpose each. An alert path that does not depend on the app's own email (owner's personal email and push).
3. **Backups (9.3):** nightly GitHub Actions job: session-pooler URL, three dumps, sanity checks (non-empty, row counts), restore into a Supabase-shaped target in the same job (weekly or every run if it is fast), write counts into the archive, `age` encrypt to the public recipient (private identity offline in two physical copies, never in GitHub), upload ciphertext as a 14-day artifact, copy ciphertext to B2, ping Healthchecks. Storage objects are exported by a separate script with a SHA-256 manifest, also encrypted before upload. Because the repo is public (D17) nothing unencrypted is ever uploaded and no counts/manifests are published as plain artifacts. Monthly: owner downloads one object from B2, decrypts offline, restores locally. Database and Storage exports are not one atomic snapshot (accepted for a few-user docs app).
4. **Email (2.14-2.18, 3.13, 3.15):** Gmail SMTP from Vercel remains the invitation path (Supabase Edge cannot use ports 25/587; Auth mail via custom SMTP is a separate Supabase service and is configured in the Auth dashboard). Quotas 3/hour and 10/day per organization, 3/day per recipient, invitation lifetime 72 h, the invite secret in the URL fragment with an explicit click, no tracking pixel, the GIF is optional decoration, the invite is fully understandable with images off, copy-link/mailto stays the fallback. A `safeMailDisplayName` sanitizer (strip CR, LF, control characters, quotes and angle brackets, cap length) is applied to the inviter name in the From display name.
5. **Exit plan (9.10):** no $0 drop-in Supabase replacement exists. Neon Free is the best database-only exit (restores plain SQL; Auth, PostgREST, Realtime and Storage would have to be rebuilt). Quarterly "exit drill": restore the latest backup into a throwaway Neon project and open the Markdown export. Static hosting exits (Netlify, Cloudflare Pages) stay as documented options only.

5. My additions beyond the report

- **Gmail account hygiene is a real trap:** app passwords need 2-Step Verification that is not "security keys only" and no Advanced Protection. A dedicated plain Gmail with an authenticator app or phone as 2SV works; then a later switch to keys-only or Advanced Protection silently breaks mail sending, and so does a Google password change (app passwords are revoked). Put both in the runbook, and make the first failed send visible in the app.
- **The pause-warning mail and the Supabase account:** the owner's Supabase account email must be one that is read; put it in the monthly checklist.
- **Backup restore doc requires** `session_replication_role = replica` and `--single-transaction` plus ON_ERROR_STOP; include in the runbook; custom role passwords are not in the dump.

- **Postgres version drift:** the dump is plain SQL and portable, but a Postgres-17-only `SET transaction_timeout` line exists in data dumps and breaks older targets; keep restore targets on 17.
- **Security of the backup path:** `age` recipient in a repository variable is public by design; the DB password and B2 key are Environment secrets; B2 application key restricted to one bucket, ideally without delete rights (UNVERIFIED which capabilities B2 offers; check when creating it); B2 bucket lifecycle rule for retention (check in the UI).
- **Test pack:** the report's 64 operations tests are kept as a checklist in `docs/22a` (lines starting at "operations test pack"); the first five to run are 1 (homepage), 14 (dump coverage of auth/storage/vault), 24-25 (storage export and SHA-256 manifest), 40 (scanner cannot accept an invite), 61 (public-repo confidentiality: nothing downloadable from CI artifacts is plaintext).

6. Open questions for the owner (recommendation first)

- **Q26** Create a new dedicated Gmail (not your main one) for sending, with authenticator-app 2SV (not security-keys-only) and recovery info you control? Recommended: yes.
- **Q27** Try Backblaze B2 sign-up now (needs email, probably a phone number; recommend, and tell me if it asks for a card)? If it asks for a card, fallback = 14-day encrypted artifact + a monthly manual download to your PC (zero signup).
- **Q28** Make two offline copies of the `age` private key (e.g. a USB stick and a printed copy) in different places? Recommended: yes; losing the key makes every backup unreadable.
- **Q29** Healthchecks.io and UptimeRobot accounts: use one throwaway-free email each and the same dedicated Gmail? Recommended: yes.

7. What changed in the master plan

Phase 9 items 9.2, 9.3, 9.8, new 9.15 (exit drill), section 13 (email setup), items 2.16-2.18 (72 h expiry, display-name sanitizer), 3.15 (limits unchanged), questions Q26-Q29 (open; no new owner decisions yet).

Audit of the ChatGPT Parts D+G report (accessibility, offline, performance, testing, engineering), 2026-10-22

Raw report: `docs/24a-RESEARCH-DG-CHATGPT-RAW.md` (about 14,700 words). Verdict: **strong and mostly right; its best calls are also the right ones for our code** (nested lists not trees, native dialog, offline = pinned read-only, no WebCrypto-in-IndexedDB, own error table, small visual-regression set). It has **three real defects** (lint stack incompatible with our TypeScript 7, wrong Speed Insights claim, tabs assumed route-backed) and left several things UNVERIFIED that I resolved. I also audited our actual code against it (section 4): that found problems the report could not see.

1. Verified TRUE (checked 2026-10-22 on primary pages or the npm registry)

Claim	Evidence
WCAG 2.5.8 Target Size (Minimum) is AA: 24x24 CSS px, exceptions spacing, equivalent control, inline, user agent, essential. A 30 px row is fine; tiny icon hit areas are not	W3C Understanding 2.5.8
3.2.6 Consistent Help is Level A (report corrected my wording; AA conformance includes A)	WCAG 2.2
1.4.11: a boundary needs 3:1 only when nothing else identifies the control; a control with visible text or a contrasting icon needs no boundary; a text input whose border is its only identifier needs the border at 3:1	W3C Understanding 1.4.11
axe-core "finds on average 57% of WCAG issues automatically"	axe-core README
WCAG 3.0 is still a Working Draft (10 Sept 2026)	w3.org/TR/wcag-3.0
APG select-only combobox example (updated 12 Aug 2025): "not intended for production environments", AT testing "essential"; Escape keeps the current value; Tab sets the value to the focused option, closes and moves on	W3C APG
CodeMirror does not bind Tab by default (keyboard-trap rule); Escape then Tab leaves the editor; Ctrl-m / Shift-Alt-m toggles tab-focus mode	codemirror.net/examples/tab
React 19.3 (2026-09-09): `ViewTransition` and Fragment Refs are stable (DOM only); no breaking-changes section; Strict Mode double-invokes effects during hydration; Trusted Types support	react.dev blog
Vite 8 default production target = Baseline Widely Available: Chrome/Edge 111, Firefox 114, Safari 16.4; `build.rolldownOptions` exists (Rolldown inside)	vite.dev/guide/build
Public repos: standard GitHub-hosted runners free; larger runners always billed	GitHub billing docs
WebKit quota: origin up to 60% of disk (browser app), overall 80%; Home Screen web apps get the same; eviction origin-wide, least-recently-used, best-effort unless persistent	webkit.org storage policy
Versions (npm registry): vitest 5.0.5, vite-plugin-pwa 2.0.0 (2026-10-03), workbox 7.4.1 (2026-05), eslint 10.13.0, @playwright/test 1.64.1, axe-core and @axe-core/playwright 4.14.0, react-aria-components **1.22.1** (the report called this UNVERIFIED: resolved, our earlier value was right), vite 8.3.5, rolldown 1.2.15, codemirror view 6.43.15	npm
METR trial: 16 experienced open-source developers, 246 tasks, AI tools made them 19% slower (July 2025, early-2025 tools). Small sample, dated: use it as a caution about verification burden, not as a current rate	metr.org

2. Wrong, overstated or missing (my corrections)

1. **ESLint 10 + typescript-eslint is NOT usable with our stack.** Our repo runs TypeScript **7.0.2** (native compiler). `typescript-eslint` 8.72.0 (and `@typescript-eslint/parser`) declare `typescript >=4.8.4 <6.1.0`; TS 7.0 ships no programmatic API; a TS 7.1 beta (npm 2026-10-20) adds one, a stable 7.1 is reported for about 24 Nov 2026 (secondary sources only) and typescript-eslint support for it is UNVERIFIED (one secondary source says the request was closed "not planned" on 7.0 GA day). The report's "ADOPT ESLint 10 + typescript-eslint" ignores this.
2. `eslint-plugin-jsx-a11y` **is stale and incompatible**: latest 6.10.2 was published 2024-10-26 (about 24 months, breaks the 18-month rule) and its peer range stops at ESLint 9; the report recommended it without checking.

3. **Vercel Speed Insights on Hobby is much weaker than the report implies.** Free tier = Real Experience Score only, path/route counts for "Great"/"Needs improvement" entries, 24 h and 7 d ranges, 10,000 events per rolling 30 days shared across the team; hitting the limit pauses collection for at least 14 days. The LCP/INP/CLS breakdown needs Plus ($10/project on Pro). So real Core Web Vitals values need our own small RUM table (`web-vitals` 6.2.3).
4. **Tabs assumption**: the report assumes the organization tabs are route-backed and says to turn them into links. In our code they are **state-backed** (`setTab`) with `role="tab"`, so they are neither deep-linkable nor a complete APG tab widget (section 4).
5. Report said our CSS tokens were unavailable and only offered a candidate palette (its contrast numbers are correct: I recomputed all nine). The real tokens are in `parts/20-styles.tsx` and I audited them (section 4).
6. UNVERIFIED items resolved: typescript-eslint 8.72.0, eslint-plugin-react-hooks 7.1.1 (supports ESLint 10), knip 6.42.1, MCP Inspector 2.12.0, pa11y 10.0.0 (2026-10-09), `@sentry/browser` 11.8.0 (versions only; Sentry's free-tier/no-card terms still UNVERIFIED). Still UNVERIFIED: screen-reader behaviour of our exact widgets, Safari 7-day storage rule on the real phone (the WebKit page does not mention it), JAWS free path, Base UI/Ariakit exact React peer ranges.
7. Minor: report's "ARIA tabs for scope chips" warning does not apply (our palette has no tab roles); report said duplicate step number 2.20 in the plan: true, now renumbered.

3. Decisions adopted (what changes in the master plan)

1. **Plan hierarchy (4.3): nested ordered lists + disclosure buttons + native checkboxes for leaf steps; no ARIA tree/treegrid in v1.** Status (blocked/deferred/cancelled) is visible text next to the checkbox, never a checkbox state; progress text "8 of 12 countable steps complete; 2 blocked", optionally `<progress>`.
2. **Behaviour layer**: keep our animated look; run a 2-day spike of React Aria (Select, Menu, ComboBox/ListBox) on the invite dropdown and the palette, adopt only if it passes our animations, native `<dialog>` and a physical Android TalkBack test. Base UI and Ariakit = assess, Radix = hold. Our Select already follows the APG select-only model (see 4), so the spike is a risk check, not a rewrite.
3. **Org tabs become route-backed** (`#/org/<id>/people` etc., links with `aria-current="page"`): fixes deep links, Back, and the broken partial tab semantics in one change. Same for the "Organizations" active/deleted switch.
4. **Diff review (5.7)**: keep `@codemirror/merge` for the visual diff, add a linear "Changes" list (hunk n of m, old/new text, native Accept/Reject buttons with context names). Do not claim the merge view is accessible until a real AT session proves it.
5. **Reorder (5.4)**: drag plus Move up / Move down / Move to in each item menu (WCAG 2.5.7), shipped together.
6. **Offline (8.3)**: explicit "Available offline on this device" pins, read-only; shell service worker (vite-plugin-pwa 2.0.0 / Workbox 7.4.1) stays registered; private IndexedDB wiped on sign-out; "Forget this device" also unregisters; never runtime-cache authenticated responses; no WebCrypto encryption of IndexedDB in v1; no offline edits; Background Sync not used.
7. **Performance budgets** (engineering policy, not standards): initial JS 180 KiB compressed target (220 KiB hard review), CSS 40 KiB, 30 requests, LCP 2.0 s, INP 150 ms, CLS 0.05 on a mid-range Android; split CodeMirror, merge, Markdown extras, import worker, labs, answer engine; virtualize only after measuring (about 300-500 rows is a profiling checkpoint). Keep Vite's default target.
8. **Testing**: Vitest 5, Playwright 1.64.1 (traces on first retry), ARIA snapshots and axe at every widget state, pinned axe version, a small visual-regression set NOW (10-20 screenshots; reverses my "deferred" 2.27), pgTAP in the local stack, Chromium PR job + Firefox/WebKit nightly, physical-device AT matrix (NVDA, Narrator, TalkBack, iPhone VoiceOver) as release gate, flake = retry-success recorded as flaky, coverage floor

70/65 with about 90% branch on pure security/version/import logic (policy, not standard), mutation testing on hold.

9. **Lint stack (9.1): Oxlint (MIT, 1.89.0, native TS/JSX, includes jsx-a11y rules and react** `exhaustive-deps`**;** `rules-of-hooks` **exists but is opt-in "pedantic") +** `tsc` **from TS 7 for types. Revisit ESLint 10 + typescript-eslint after it supports TS 7.1.** Rule sets to enable are TRIAL: verify on our code. Biome 2.6.2 is the alternative (a11y/hooks rule coverage not verified here).
10. **Error + RUM (9.8)**: first-party `client_error_events` (release, route template, sanitized fingerprint, coarse browser) and a `web-vitals` RUM table (route template, release, metric, value, device class; 30 days; no ids, titles, queries, hashes, secrets). Sentry/GlitchTip on hold (terms unverified). Speed Insights optional only as a rough score.
11. **Browser baseline (Q21 recommendation: yes)**: current + previous Chrome/Edge/Firefox/Safari including iOS Safari; keep Vite's default target; adopt dialog, popover (selectively), `:has()`, container queries, `color-mix()`; trial `@starting-style`, View Transitions; hold anchor positioning, scroll-driven animation, Temporal and customizable `<select>` as dependencies.
12. **Auth screens (3.13/3.14)**: one OTP field with `autocomplete="one-time-code"` and paste, `new-password`/`current-password` autocomplete, paste never blocked, manual TOTP seed plus QR, Turnstile test keys in CI and a documented fallback, second TOTP factor before enforcing AAL2. Passkeys: assess only (Supabase marks them experimental).
13. **Assistant rules (AGENTS.md)**: bounded changes with acceptance tests; no package installed until the owner verifies registry name, maintainer, age, licence and need; no unreviewed migrations, security policy or deploys; no secrets, dumps or share links in prompts.

4. Audit of OUR code against the report (new, from `parts/*.tsx` and the CSS tokens)

Already good (keep): native `<dialog>` with `showModal()` in 5 dialogs (form modal, confirm, import, palette, settings); popups are portalled INTO the nearest open dialog (`popHost`), which avoids the top-layer/popover pitfall the report warns about; custom Select is a select-only combobox (`role=combobox`, `aria-activedescendant`, `aria-controls`, Home/End, typeahead, Escape keeps the value, active option scrolled into view); menus use `menuitem` buttons; motion setting (system or forced) drives `data-motion`; toast is `role=status` and pauses on hover; the views I checked have an `<h1>`; `nav` landmarks have labels; segmented controls use `aria-pressed`; diff has a `forced-colors` rule; native focus ring `2px solid #6da8ff` (7.0:1) on `:focus-visible`; palette scopes are not tabs; text tokens pass (see below). **Problems found:**

1. **Org tabs** (`parts/49-orgs.tsx` lines 231, 369): `role=tablist/tab` with `aria-selected` only: no roving tabindex, no arrow keys, no `aria-controls`/`tabpanel`, state-only. Every tab is a Tab stop and arrows do nothing: worse than plain buttons. Fix = route-backed links (decision 3).
2. **No skip link, no per-route** `document.title`**, no focus/announcement on route change** (title is only touched for the unread-count prefix; `hashchange` handler does not move focus). One `<main>` exists.
3. **Control borders fail 1.4.11 where the border is the only identifier**: `--line` #2a2926 and `--line2` #3b3a35 give only 1.04-1.62:1 against our surfaces (needs 3:1). Applies to text inputs, textareas, search fields (placeholder alone is weak). Buttons and selects with text/icons are exempt. Candidate control-border token `--ctl-line: #77756c` = 3.99:1 on `--bg`, 3.88 on `#181715` (textarea fill), 3.69 on `--raise`, 3.28 on `--bubble`; check on `--hover`/`--active` before adopting.
4. **Select**: `aria-controls` points at the inner wrapper `div` (`${id}-l`) that has no role; the `role="listbox"` is on the `.pop` element. Put id and role on the same element. Tab currently closes without committing the active option (APG commits): decide and test, minor.
5. **forced-colors** handled only for the AI diff; no global pass (borders as the only state cue, selected/active rows, focus ring) yet.

6. **Contrast of accent**: `--acc` text on `--active` is 4.48:1 (just under 4.5); primary button text #141413 on `--accb` is 4.73:1 (passes). `--tx4` on `--active` 4.90 passes. All other text/semantic tokens pass on every surface (`--tx3` 6.1-8.1, `--blue` 5.2-6.9, `--ok` 7.8-10.3, `--bad` 5.8-7.6, `--warn` 7.3-9.7, `--focus` 5.8-7.6).
7. **Hit areas**: avatar 26 px and dense `.btn` 26 px pass; `.nav-n` 20 px is a non-interactive badge. I did not measure every icon button: add a Playwright geometry test that fails any visible interactive element under 24x24 without the spacing exception.
8. **No linter exists** (no ESLint/Oxlint config): adding Oxlint is the cheapest accessibility gain (static `jsx-a11y` checks).
9. Plan items already numbered twice (`2.20`): second one renumbered 2.20b.

5. New plan steps created by this audit

2.40 skip link + route title + route focus manager; 2.41 route-backed org tabs; 2.42 control-border token + a contrast test script in CI that parses the CSS tokens and asserts ratios (cheap, deterministic); 2.43 global `forced-colors` pass; 2.44 Playwright target-size geometry test; 2.45 accessibility statement and issue template; 2.46 AT release gate; the Select id/role fix and Tab-commit decision are folded into 2.7; 9.16 Oxlint + `tsc`; 9.17 `web-vitals` RUM table and client error table (merges with 9.8); 9.18 AGENTS.md rules. Modified: 2.7, 2.8, 2.11, 2.12, 2.20, 2.26 (also 320 px), 2.27 (small set now), 4.3, 5.4, 5.7, 8.3, 9.1, 9.7, 9.8, 9.11.

6. Open questions for the owner (recommendation first)

- **Q30** Make the organization tabs real routes (`#/org/<id>/people`)? Recommended: yes.
- **Q31** Run the 2-day React Aria spike (Select, Menu, ComboBox) before deciding, or keep and fix our own widgets only? Recommended: spike, because a physical Android TalkBack test is needed either way.
- **Q32** Do you own an Android phone for TalkBack and a Windows PC for NVDA/Narrator (free)? Recommended to confirm; otherwise screen-reader claims stay UNVERIFIED.
- **Q33** Offline: only pinned read-only documents (recommended), or none for now?
- Q21 (browser baseline) answered by the report's evidence: recommend yes.